REYNOLD FREGOLI

Department of Mathematics
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Academic Appointments

Assistant Professor 2026–present
North Dakota State University

Postdoctoral Assistant Professor 2023–2026
University of Michigan
Mentor: Prof. Ralf Spatzier

Postdoctoral Researcher 2021–2023
University of Zürich
Mentor: Prof. Alexander Gorodnik

LMS Early Career Fellow 2020–2021
University of York
Mentor: Prof. Victor Beresnevich

Education

Ph.D. in Mathematics 2020
Royal Holloway University of London
Advisor: Prof. Martin Widmer

M.S. in Mathematics 2016
University of Pisa
Advisor: Prof. Ilaria Del Corso

B.S. in Mathematics 2014
University of Pisa
Advisor: Prof. Ilaria Del Corso

Research Interests

Ergodic theory, homogeneous dynamics, Diophantine approximation, analytic number theory.

Publications

1. J. Cheng, R. Fregoli, and B. Guo, *Higher dimensional moving averages and submanifold-genericity*, ArXiv:2507.15498 (2025), to appear in Discrete Contin. Dyn. Syst.

2. M. Björklund, R. Fregoli, and A. Gorodnik, *A uniform metrical theorem in multiplicative Diophantine approximation*, Forum. Math. Sigma **14** (2026), e39.

3. R. Fregoli and D. Kleinbock, *On multiplicatively badly approximable vectors*, J. Number Theory **278**

(2026), 570-621.

4. M. Björklund, R. Fregoli, and A. Gorodnik, *Decorrelation estimates for translated measures under diagonal flows*, J. Mod. Dyn. **25** (2025), 401–441.

5. R. Fregoli, *A remark on the set of exactly approximable vectors*, Proc. Amer. Math. Soc. **152** (2024), 3177-3182.

6. R. Fregoli and C. Zheng, *A shrinking-target problem in the space of unimodular lattices in the three-dimensional Euclidean space*, Sel. Math. (N.S.) **30** No. 57 (2024).

7. R. Fregoli, *Multiplicatively badly approximable matrices up to logarithmic factors*, Math. Proc. Cambridge Philos. Soc. **172** No. 3 (2022), 685-703.

8. R. Fregoli, *A note on bounded exponential sums*, Bull. Lond. Math. Soc. **53** No. 2 (2021), 416-425.

9. R. Fregoli, *On a counting theorem for weakly admissible lattices*, Int. Math. Res. Not. **2021** No. 10 (2021), 7850-7884.

10. R. Fregoli, *Sums of reciprocals of fractional parts*, Int. J. Number Theory **15** No.4 (2019), 789-797.

Preprints[1]

1. R. Fregoli, T. Kim, D. Kleinbock, V. Nekrasov, and A. Rao, *Well approximable Dirichlet spectrum and full Hausdorff dimension of the Folklore Set* (in preparation).

2. R. Fregoli and Y. Zhang, *Diophantine properties of two-dimensional Mahler vectors* (in preparation, click here for a preliminary version).

3. P. Bandi and R. Fregoli, *On the Hausdorff dimension of weighted exactly approximable vectors*, arXiv:2601.12121 (2026).

4. M. Björklund, R. Fregoli, and A. Gorodnik, *Central limit theorems in multiplicative Diophantine approximation*, ArXiv:2601.13726 (2026).

5. P. Bandi, R. Fregoli, and D. Kleinbock, *Submanifold-genericity of $\mathbb{R}^d$-actions and uniform multiplicative Diophantine approximation*, ArXiv:2504.02258 (2025).

6. R. Fregoli (Appendix by M. Björklund, R. Fregoli, and A. Gorodnik) *Sums of reciprocals of fractional parts II*, ArXiv:2304.02566 (2023).

Invited Talks

- The Hausdorff dimension of the Folklore Set, AMS Joint Mathematical Meetings, Chicago (US), Jan 2027
- Orbits, chaos, and rational numbers, Colloquium at North Dakota State University, Fargo (US), Mar 2026.
- Ergodic theorems for dilates of manifolds, New England Dynamics and Number Theory Online Seminar (organized by D. Kleinbock, F. Ramirez, and H. Li), Nov 2025.

[1]The second preprint includes two lemmas obtained with the assistance of artificial intelligence. AI was not used in the remaining manuscripts.

- Ergodic theorems for dilates of curves and applications to Diophantine approximation, Research seminar at the University of Illinois, Chicago (US), Nov 2025.
- Ergodic theorems for dilates of curves and applications to Diophantine approximation, Research seminar at TU Graz, Graz (Austria), Jul 2025.
- Exact Diophantine approximation in the weighted setup, Workshop on Diophantine Approximation and Related Fields, University of York, Heslington (UK), Jun 2025.
- Equidistribution of geometric sequences on nilmanifolds, Research seminar at Brandeis University, Waltham (US), Feb 2025.
- Equidistribution of geometric sequences on nilmanifolds, Research seminar at Ohio State University, Columbus (US), Jan 2025.
- Central limit theorems and higher-rank diagonal actions, Workshop on Dynamical Systems and Related Topics, Pennsylvania State University, State College (US), Nov 2024.
- Central limit theorems and higher-rank diagonal actions, Research seminar at the University of Michigan, Ann Arbor (US), Nov 2024.
- Improvements to Dirichlet's theorem in the multiplicative setup and equidistribution of averages along curves, Diophantine Approximation, Fractal Geometry, and Related Topics, Université Gustave Eiffel, Marne la Vallée (France), Jun 2024 (contributed).
- Some open problems in homogeneous dynamics, Capsule talk at the University of Michigan, Ann Arbor (US), Sep 2023.
- On the Dani corrspondence and its applications, Simons Semester on Dynamics (3 lecture mini-course), Banach Center, Warsaw (Poland), May 2023.
- Almost-sure estimates for sums of reciprocals of fractional parts, Online Seminar in Diophantine Approximation and Related Topics (organized by A. Marnat and N. Moshchevitin), May 2023.
- Shrinking-target problems and the injectivity radius function, Research seminar at Scuola Normale Superiore, Pisa (Italy), Apr 2023.
- Multiplicatively badly approximable vectors, Ergodic Theory and Dynamical Systems Conference, ICTS Bangalore (India), Dec 2022.
- Multiplicatively badly approximable vectors, Research seminar at the University of York, Heslington (UK), Nov 2022.
- Multiplicatively badly approximable vectors, Online Seminar in Diophantine Approximation and Related Topics (organized by A. Marnat and N. Moshchevitin), Oct 2022.
- Multiplicatively badly approximable vectors, Research seminar at University of Zürich, Zürich (Switzerland), Oct 2022.
- Randomness and quantitative aspects of the Littlewood conjecture, Summer School on Group Actions, Ohalo (Israel), Jul 2022.
- Randomness and quantitative aspects of the Littlewood conjecture, Research seminar at Shanghai Jiao Tong University, online, Mar 2022.

- Multiplicatively badly approximable matrices and the Littlewood conjecture, Online Seminar in Diophantine Approximation (organized by N. Moschevitin), July 2020.
- Bounded exponential sums, Research seminar at the University of Bristol, Bristol (UK), March 2020.
- Bounded exponential sums, Research seminar at the University of York, Heslington (UK), January 2020.
- Bounded exponential sums, Research seminar at TU Graz, Graz (Austria), Jan 2020.
- Bounded exponential sums, Research seminar at Royal Holloway University of London, Egham (UK), October 2019.
- Sums of reciprocals of fractional parts, Baikal Number Theory Conference, Ol'khon Island (Russia), Aug 2019 (contributed).
- Sums of reciprocals of fractional parts, Research seminar at the University of York, Heslington (UK), May 2019.
- Sums of reciprocals of fractional parts, Research seminar at the University of Manchester, Manchester (UK), May 2019.
- Diophantine approximation: a new application of weak admissibility, Research seminar at Royal Holloway University of London, Egham (UK), Nov 2018.
- Sums of reciprocals of fractional parts, Research seminar at Royal Holloway University of London, Egham (UK), Apr 2018.
- Sums of reciprocals of fractional parts, Research seminar at Royal Holloway University of London, Egham (UK), Feb 2018.
- On some algebraic properties of the locally free class group, Research seminar at the University of Pisa, Pisa (Italy), Apr 2017.
- An application of Frölich's Hom Description, Research seminar at the University of Pisa, Pisa (Italy), Mar 2017.

Teaching Experience

North Dakota State University
Math 166 – Calculus II (Fall 2026).

University of Michigan
Math 425 – Introduction to Probability (Winter 2024, Fall 2025, Winter 2026).
Math 115 – Calculus I (Fall 2024).
Math 116 – Calculus II (Fall 2023, Winter 2025).

University of Zürich
Mat 132 – Analysis II for Physics (Spring 2023)
Student seminar supervisor (topic: Fourier Analysis) (Fall 2022).
Teaching assistant for Complex Analysis (Spring 2022).
Teaching assistant for Analysis III (Fall 2021).

Awards and Grants

- LMS Early Career Fellowship, London Mathematical Society, 2021.

Academic Service

- Referee for Int. J. Number Theory, Discrete Contin. Dyn. Syst., J. Reine Angew. Math., Proc. Edinb. Math. Soc., Int. Mat. Res. Not., Adv. Math., J. Lond. Math. Soc., Selecta Math. (N.S.), Ark. Mat..
- Conference organizer: Egham-Reading-London Arithmetic Statistics Seminars (ERLASS), Royal Holloway University of London, first seminar (Dec 2019).
- Seminar organizer: Ergodic Theory and Dynamical Systems Seminar, joint University of Zürich and ETH Zürich, Sep 2022 - May 2023.